AeroSentinel: Autonomous CubeSat Fault-Recovery RL Environment

System Design Document & Verification Specification | Devesh Srivastava (Avionics Head, IIT BHU Rocketry)

1. System Overview & Physical Grounding

AeroSentinel evaluates autonomous AI agents tasked with diagnosing and remediating in-flight anomalies on a 3U CubeSat in Low Earth Orbit (500 km, 95-minute period). Unlike traditional coding sandboxes that evaluate static string matches or unit tests visible in the repository, AeroSentinel embeds an **authoritative orbital multi-physics engine**. The satellite's state is continuously evaluated across sunlight and eclipse cycles: solar array generation (0.06 m², GaAs triple-junction), battery Coulombic state-of-charge (40.0 Wh Li-ion, 3.10V cutoff), nonlinear bus voltage dynamics, 3-axis reaction wheel angular momentum and attitude precession, thermostat hysteresis thermal balance, and UHF 437.5 MHz Friis link budget margins with 141.5 dB free-space path loss. All engineering parameters are derived from embedded flight software practices developed for rocketry avionics.

2. Flight Task Suite (Zero-Leakage Specification)

Tasks provide realistic symptoms, telemetry logs, and flight manuals (docs/flight_manual.md), completely eliminating answer parameter leakage. The agent must diagnose root causes from telemetry and calculate repairs:

Task ID	Difficulty	Anomaly Symptoms & Physics Dynamics	Nominal Solution Criteria
`eps_eclipse_recovery`	Easy	Bus voltage collapses below 3.0V in 35-min eclipse; payload brownout.	Recalibrate GaAs MPPT solar efficiency to 0.72 & safe undervoltage cutoff >= 3.20V.
`adcs_detumble_pointing`	Easy	Tumbling at >3 deg/s; continuous off-sun precession starves solar panels.	Zero reaction wheel momentum bias; set pointing deadband to 0.05 deg.
`thermal_battery_preservation`	Medium	Battery temperature climbs toward 50°C runaway under stuck heater setpoint.	Reconfigure thermostat setpoint to 15°C and emergency cutoff to 45°C.
`rf_downlink_closure`	Medium	Zero telemetry packets received during 8-min pass; link margin negative.	Configure 30 dBm TX, 8 dBi antenna, dynamically query 141.5 dB path loss.
`safe_mode_cascade`	Medium (Seeded)	Compound safe-mode tumble with low battery and multi-subsystem drift.	Coordinated multi-subsystem parameter repair across EPS, TCS, ADCS, COMMS.

3. Unbreakable Isolation & Security Defenses

AeroSentinel is hardened against adversarial model behaviors, sandbox escapes, and scoring manipulation:
• **Authoritative Unified State & Latched Violations**: Virtual time, Coulombic battery energy, and thermal equilibrium form a single authoritative trajectory. Any irreversible catastrophic violation (bus collapse <= 3.10V, runaway > 50°C) is permanently latched.
• **Isolated Unprivileged Evaluation Worker**: Untrusted agent Python code is NEVER executed in the harness container. Behavioral checks execute via /eval_script in isolated ephemeral directories under unprivileged sandbox-agent (UID 1001).
• **Dynamic Secret Token Authorization**: Privileged endpoints (/reset, /state, /fetch_configs, /eval_script) require a dynamically generated 256-bit token created at container startup into a root-owned mode 0600 volume.
• **Physical Channel Law Throughput**: Downlink byte throughput is bounded by configured baud rate (9600 bps -> max 576 kB in 8 min).
• **Monotonic Execution Deadlines**: A global monotonic clock bounds model calls, 429 retries, and tool calls, preserving a 15-second verification buffer to ensure every run completes strictly within the 600-second contractual budget.

4. Programmatic Outcome-Based Reward Design

Scoring is entirely programmatic (zero LLM-as-judge) and evaluates physical mission success over a 95-minute orbit:
Reward = Power_Balance (0.35) + Thermal_Stability (0.25) + ADCS_Precision (0.20) + Comms_Closure (0.20)
• **Catastrophic Zeroing**: Irreversible failure (bus collapse <= 3.10V, thermal runaway > 50°C, freezing < -15°C) strictly clamps the total reward to 0.00.
• **Continuous Partial Credit**: Sustained battery SoC, settled pointing angle within deadband, and downlink packet throughput yield proportional credit for genuine engineering progress.

5. Known Weaknesses & Hardening Defenses

Known Vulnerability / Threat	Architectural Defense Implemented
Agent attempts to inspect or tamper with verifier files	Verifier runs entirely in the separate Harness container; sandbox filesystem has zero grading code.
Untrusted agent Python code executed in harness	All script evaluation runs via /eval_script inside the sandbox container under UID 1001 sandbox-agent.
Agent submits comment-only bypass for link budget	Dynamic behavioural verification executes calc_margin() against 3 randomized test fixtures with varying path loss.
Agent submits infinite loops (while True: pass) or memory hogs	Subprocess kills calls at 20 seconds; container enforces 512MB RAM and 256 PID limits.
Symlink or path traversal attack (../../etc/passwd)	safe_path() canonicalizes path via resolve() and checks relative_to(WORKSPACE), blocking /proc, /run, /etc.
Provider rate limit / context overflow exceeds 600s	Monotonic deadline tracks remaining time across API calls, 429 backoff retries, and tools, reserving 15s for verifier.

6. Experimental Baselines Summary

Oracle Baseline: Scores **1.00 / 1.00** across all 5 tasks (Threshold: >= 0.95). **No-Op Baseline**: Scores **0.00 / 1.00** across all 5 tasks (Threshold: <= 0.05). Total gate execution duration: **0.05 seconds** (Budget: 120s). Live model gemini-3.5-flash-lite autonomously achieved **1.00 / 1.00** across multiple tasks using interactive orbital telemetry.